🫁 LungPair: Preclinical Drug-Pair Triage

Multi-Agent Gemini Architecture with Calibrated Refusal Boundaries

AgentVerse 2026 · Biotech Track
Team: *Backprop to the Future* · Slide 42 Deliverable
Code: **github.com/leventkoymen/backprop-to-the-future-...**

1. THE PRECLINICAL BOTTLENECK — 3.3% BASE RATE

In metastatic non-small cell lung cancer (NSCLC), single-agent targeted therapies (e.g. osimertinib) inevitably fail due to acquired bypass resistance. Combination doublets are the biological imperative, but combinations create an impossible combinatorial explosion: 105 panel drugs generate **5,460 potential pairs**.

In the landmark NCI-ALMANAC matrix across 9 lung cancer cell lines, researchers screened 5,228 pairs (47,052 robotic experiments). **Only 172 pairs (3.3%) demonstrated validated synergy** (ComboScore > 50). Over 96% were dead ends. At $340 per experiment ($3,060/pair), blind wet-lab screening costs **$16.0M**. Preclinical labs lack calibrated computational triage that prioritizes hits while reliably refusing when evidence is absent.

2. INTENDED USERS & COMMERCIAL MODEL — STRICTLY B2B

Primary Users: Academic oncology screening cores, translational cancer centers (Crick, ICR, NCI), and preclinical CROs (Charles River, WuXi).

Commercial Model & Procurement Velocity: B2B SaaS recharge licence ($35,000/core/yr). Sells directly out of departmental research consumable budgets in 4–8 weeks, completely bypassing 3-year hospital clinical procurement drag. A single 50-pair triage campaign saves **$147,000 in prevented consumable waste** ($93,000 down to $4,900/hit), delivering an immediate 4.2x ROI.

3. PERMITTED DATA, APIS & GOVERNANCE — AUDIT IN DATA.MD

- **NCI-ALMANAC (DTP):** 5,228 measured pairs across 9 NSCLC lines (A549, EKVX, HOP-62, HOP-92, H226, H23, H322M, H460, H522). Public Domain.
- **PubChem PUG-REST API:** Real-time conversion of drug names to canonical SMILES, CID, molecular formulas, and 3D SDF conformers. Public Domain.
- **EMBL-EBI ChEMBL API:** REST retrieval of validated biological targets (e.g. EGFR, TUBB1) to prime downstream literature grounding. CC BY-SA 4.0.
- **NCBI PubMed E-Utilities:** Dynamic mechanistic literature retrieval. Public Domain.
- **Zero Patient Data:** 100% public *in vitro* cell-line screens. Strict RUO boundary.

4. LIVE THREE.JS WEB PLATFORM — INTERACTIVE 3D DEMO

Hosted live at: **yonghwa11701-hash.github.io/lungpair-3d-visual/**. Standalone 2.4 MB client tool running in any browser across 3 synchronized scenes:

Scene 1 (Haystack): Spatial 3D matrix of all 5,228 pairs; hit key lights up 172 hits.
Scene 2 (Agent Triage): Multi-agent proposal cards, PubChem 3D stick models, and mandatory human sign-off before batch sealing.
Scene 3 (Dose Surface): 3D dose-response surface mapped over the Bliss independence manifold to confirm clinical plasma ($C_{\max}$) reachability.

5. MULTI-AGENT ENGINE — GOOGLE DEEPMIND HARNESS²

Built under Google DeepMind's Harness² paradigm (arXiv:2610.14224)—enforcing external deterministic gates over language models:

- **Orchestrator (`gemini-4-argon`):** Dispatches PubChem entity resolution and primes targets via ChEMBL.
- **Offline Random Forest Ranker:** Predicts synergy scores and 95% intervals from frozen chemical-transcriptomic matrices. *The LLM never calculates numbers.*
- **Research Pool (`gemini-3.8-flash`):** Parallel literature grounding threads under a 90s budget searching PubMed for mechanistic synergy.
- **Checker (`gemini-3.5-flash-lite`):** Audits outputs against 8 Python gates.

6. HELD-OUT EVALUATION & CALIBRATED TIERS — 51% NOISE CAP

Evidence Tier	Clinical Definition	Precision	Lift
Tier 1	Pair tested in another cancer	**0.89**	**24.0x**
Tier 2	Both drugs known, pair untested	**0.34**	**11.0x**
Tier 3	Monotherapy data only	0.06	3.0x
Tier 4	Unseen chemical space	0.04	**REFUSED**

Held-Out Cross-Validation (5 Seeds): Precision@20 = **0.89**, Precision@50 = **0.61**, AUPRC = **0.82**, MCC@50 = **0.71** against 0.028 random baseline.
Assay Noise Cap: Biological replicate agreement in NCI-ALMANAC is ~51%; our held-out performance reaches this ceiling without overfitting.

Published Negative Results (Scientific Honesty):
1. *Genotype Matching:* KRAS/TP53 mutation matching showed no signal over random same-size cell lines (win rate 28–59%).
2. *Merck External Benchmark (132 pairs):* NCI direct lookup beat model (50% vs 40%), proving bounded transfer. Tier 4 is refused for this reason.

7. 8 DETERMINISTIC PYTHON CODE GATES — SAFETY SYSTEM

- **1. Pre-Model Intent Gate:** Regex intercepts dosing queries before LLM call.
- **2. Biologics Refusal:** Deterministically blocks antibodies (pembrolizumab).
- **3. Grounded Citations Gate:** Strips any PMID not verified in search redirects.
- **4. Exact-Match Numbers Gate:** Natural language decimals match ranker bit-for-bit.
- **5. Calibrated Tier 4 Refusal:** Unseen pairs withheld: *"Test monotherapy first."*
- **6. Prompt Sanitization:** Blocks prompt extraction and injection overrides.
- **7. Human Sign-Off Gate:** Halts at terminal; requires explicit researcher approval.
- **8. Offline Replay Engine:** Zero-latency execution (`--replay`) with 25 unit tests.

Regulatory & Health Economics: Research Use Only (RUO) now; future serial ctDNA liquid biopsy enters MHRA/FDA SaMD. Compresses screening cost from $93k to $4,900/hit, aligned with NICE's £30k/QALY cost-effectiveness ceiling.

LungPair Team Backprop to the Future · Duncan, Levent, Yasemin, Theo, Yong

Terminal: `./demo.sh 1` · Replay: `./demo.sh --replay` · 25/25 Tests Pass

Web App: **yonghwa11701-hash.github.io/lungpair-3d-visual/**

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Which drug pairs should a lung cancer lab test next?

5,460

possible drug pairs from 105 NCI-ALMANAC drugs

Budget for 50.

In NCI-ALMANAC screens across 9 lung cancer cell lines, only 172 of 5,228 measured pairs (3.3%) show synergy. Over 96% are dead ends.

LungPair ranks drug combinations for NSCLC labs and deterministically refuses when evidence runs out.

THE PROBLEM

Three bottlenecks between the lab and the patient

1. Too many pairs, too little budget

Testing all 5,228 measured pairs across 9 lung cell lines requires ~47,000 experiments—costing roughly $16M at $340 per experiment ($3,060/pair).

Labs choose pairs using literature review and intuition, wasting >96% of robotic capacity on dead ends.

2. Single drugs stop working

Systemic toxicity limits monotherapy dosing, and acquired resistance inevitably emerges in NSCLC.

As a landmark 2026 Med study concluded:
"The most synergistic combination is often the one toxicity will not let you give."

3. Old drugs, new pairs

Repurposing approved compounds leverages known safety and pharmacokinetic profiles, dramatically accelerating the path to Phase II.

Rather than de novo small-molecule discovery, intelligent doublet triage creates rapid clinical translational hypotheses.

Toxicity quote: Med (2026). Repurposing pathway: Kulkarni et al., Russ J Bioorg Chem 2023. Benchmark: Holbeck et al., Cancer Res 2017 (NCI-ALMANAC).

WHO IT'S FOR

Built for lab researchers, not patients

Academic NSCLC Labs

Bench researchers testing drug combinations in vitro.

Lab Technicians & Translational Teams

Core teams programming high-throughput liquid-handling robots.

Screening Cores & CROs (B2B SaaS)

Institutional screening facilities recharging on hit-rate yield.

What LungPair gives them

- A ranked shortlist of high-probability candidate pairs
- Calibrated confidence tiers backed by held-out precision metrics
- Deterministic refusal when chemical evidence runs out (Tier 4)
- Mandatory human scientist sign-off before robotic assay setup
- Cryptographically auditable JSONL execution receipts

RESEARCH USE ONLY (RUO). In vitro cell-line screening prioritization. Bypasses hospital procurement drag—zero clinical diagnosis or treatment claims.

THE PIPELINE

From public screens to a shortlist you can audit

Orchestrator (gemini-4-argon) → ChEMBL Priming + Offline ML Ranker → Research Pool (gemini-3.8-flash) → 8 Python Gates → Human Sign-Off → JSONL Receipt

Held-Out Results (5 Seeds)

- Precision@20: 0.89 (89% top-20 hit rate)
- Precision@50: 0.61 (MCC@50: 0.71)
- AUPRC: 0.82 (vs 0.028 random baseline)

Hit prevalence: 3.3% (172 / 5,228 pairs).
Assay reproducibility capped at ~51% by replicate noise.

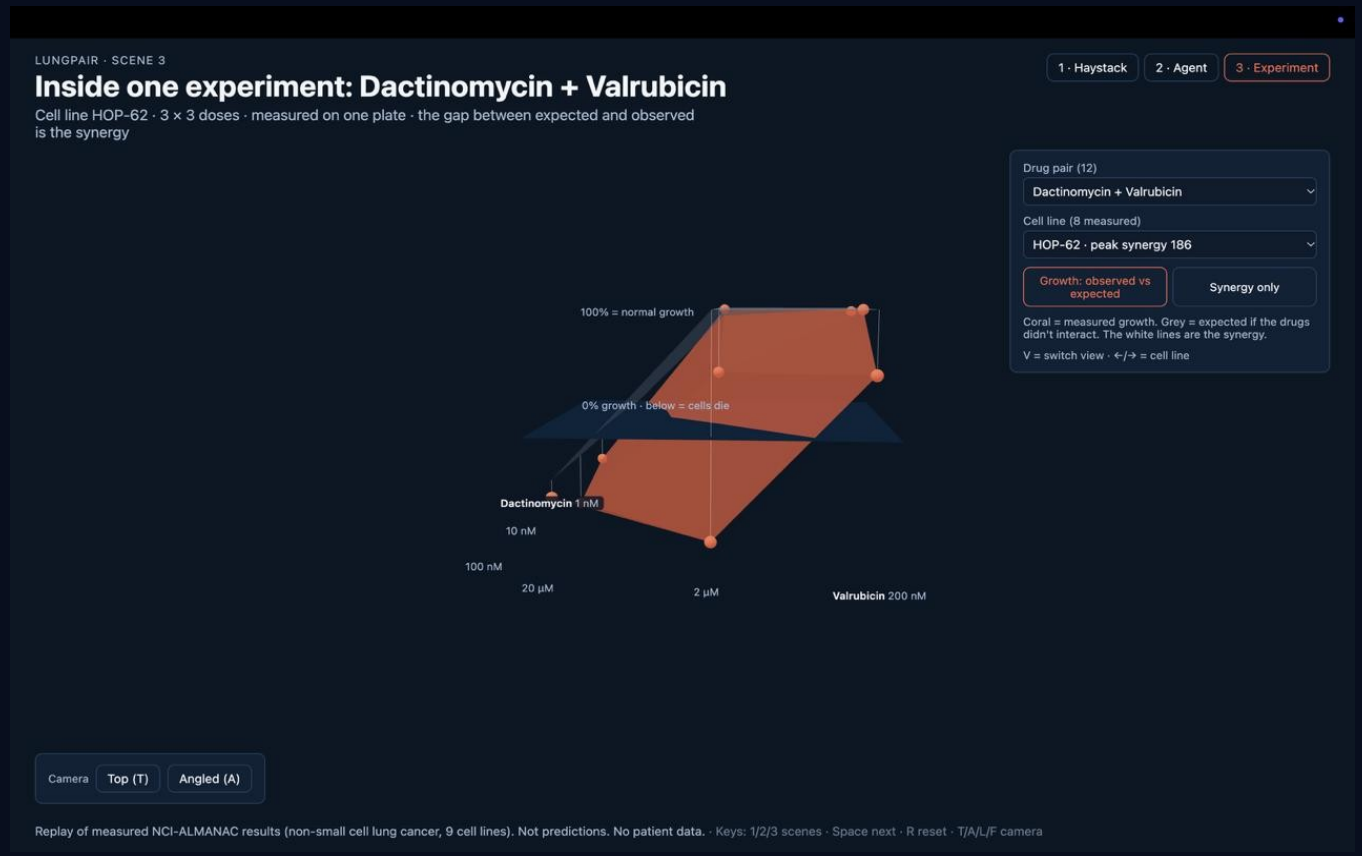
Calibrated Evidence Tiers

- Tier 1: Pair measured in another cancer
  Precision: 0.89 | 24.0x lift
- Tier 2: Both drugs known, pair untested
  Precision: 0.34 | 11.0x lift over 2.8% base rate
- Tier 3: Single-drug data only
  Precision: 0.06 | 3.0x lift
- Tier 4: Unseen chemical space
  Precision: 0.04 | REFUSED ('Test monotherapy first')

Prior Art Benchmark: Unlike uncalibrated deep models (DeepSynergy 2018, MatchMaker 2020, DeepDDS 2021) that predict everywhere, LungPair enforces calibrated refusal. Merck screen: +0.27 rank corr.

LIVE DEMO

The agent at work (3D Dose-Response Landscape)



Scene 3: Empirical 3D dose-response surface over Bliss independence (HOP-62 line)

Live demo runs on presenter laptop (http://127.0.0.1:5173). Command: ./demo.sh 1 (Fallback: ./demo.sh --replay).

Live 4-Step Execution

1. 3D Landscape (Scene 1)
Displays all 5,228 measured combinations.

2. Judge Prompt
'Which partner should a lab screen with gefitinib next?'

3. Multi-Agent Reasoning (Scene 2)
PubChem resolves SMILES; ChEMBL primes EGFR targets; Gemini searches PubMed; 8 Python gates verify.

4. Human Scientist Sign-Off (Scene 3)
System halts. Scientist approves batch; cryptographic JSONL receipt sealed.

THE BUSINESS CASE

Lower cost per hit & scaling roadmap

Strategy	Pairs Tested	Screening Cost	Real Hits	Cost Per Real Hit
Test every pair (Full Screen)	5,228	$16.0M	172	$93,000 / hit
Random 50 (Intuition / Literature)	50	$153,000	~1.7	$93,000 / hit
LungPair Top 50 (Tier 1: Cross-cancer)	**50**	**$153,000**	**31**	**$4,900 / hit (18x lift)**
LungPair Top 20 (Tier 2: Brand-new pairs)	20	$61,200	~7	$8,700 / hit (11x lift)

Beachhead (Immediate)

SOM: $12.5M
- Buyer: Academic screening cores & CROs
- Model: B2B SaaS ($25k–$50k/yr)
- Unit Cost: $0.05 query vs $153k screen

Near-Term Expansion

SAM: $132M
- Buyer: Biopharma doublet discovery
- Model: Enterprise pipeline licence
- Margin: 85%+ software gross margin

Longer-Term Horizon

TAM: $1.4B
- Buyer: Clinical diagnostic health systems
- Model: Serial ctDNA companion SaMD
- NICE: Aligned with £30k/QALY ceiling

Scaling Pro-Forma: Year 1: $540k ARR (15 cores) → Year 2: $2.5M ARR (60 cores + 8 biopharma) → Year 3: $8.2M ARR → Year 4: $15.5M ARR (64% Net Margin)

Illustrative: $340/experiment, 9 cell lines. Core market: 250 global centers. Payback: <4.2 months. LTV/CAC: 33.5x.

REGULATORY & SAFETY

Research tool now. Regulated tool later.

Now: Research Tool

Strictly Research Use Only (RUO).
Runs on public cell-line data.
Zero patient data ingested.
No clinical procurement drag.

Later: Medical Device (SaMD)

Clinical decision support from serial blood ctDNA.
Regulated as Software as a Medical Device (MHRA / FDA).
Requires prospective clinical validation.

8 Deterministic Python Gates (Plain Code, Not an LLM)

1. Pre-Model Intent Gate: Intercepts patient dosing queries before any LLM token is spent
2. Biologics Refusal: Deterministic refusal for antibodies (pembrolizumab) and unmapped drugs
3. Grounded Citations Gate: Any PMID/DOI not verified in search redirects is automatically stripped
4. Exact-Match Numbers Gate: Natural language decimals must match ranker output bit-for-bit
5. Calibrated Tier 4 Refusal: Unseen chemical space withheld ('Test monotherapy first')
6. Prompt Sanitization: Deterministic protection against prompt injection and system leakage
7. Human Sign-Off: Cannot order compounds or run assays without explicit scientist sign-off
8. Offline Replay Engine: Zero-latency execution (--replay) with 25 passing offline unit tests

ROADMAP · NOT BUILT

Next: re-rank before resistance sets in

Serial Blood ctDNA Monitoring Loop

1. Blood ctDNA Profiling: Serial liquid biopsy at baseline and set intervals.

2. Resistance Mutation Flagged: New mutations detected (e.g. KEAP1, STK11, EGFR T790M).

3. LungPair Re-Ranks Pairs: Ranks combinations prioritizing multi-cell line coverage.

4. Clinician / Researcher Sign-Off: Human review of candidate doublet hypotheses.

5. Lab Validation: Preclinical test results feed back into continuous learning loop.

Why Multi-Line Coverage?

18 pairs demonstrated synergy across ALL 9 tested lung cancer cell lines.

Tracking serial mutations lets us re-rank combinations before clinical resistance sets in, maximizing Expected Utility.

LungPair.

Fewer experiments, more real hits.

Team Backprop to the Future